INTER-IIT INTERNAL HACKATHON

Optimal Fantasy Cricket Team:

Availability-Aware Scoring and Exact Constrained XI Selection

Final Model v3 — Technical Report

Methodology, experiments, quantitative metrics, model explainability, selection-algorithm justification and leakage audit

Headline result

A single gradient-boosted point model raises the top-11 capture ratio from **57.16%** to **66.63%** (+9.47 percentage points) and cuts MAE by **32.8%** against a last-5 baseline. The pipeline is auditable and leakage-safe, and all data used is dated on or before 31 December 2025.

Abstract

Fantasy cricket selection is often framed as a ranking problem: predict each player's points and pick the top eleven. In practice this framing fails for two reasons. First, an announced squad is not the playing XI, so a player with a high expected score who does not take the field contributes nothing. Second, the chosen eleven must satisfy hard role and team constraints that a ranking model cannot guarantee.

This report presents Final Model v3, a system that separates the two concerns. A statistical layer predicts fantasy points for every candidate with a single histogram gradient boosting regressor trained with MAE loss on 22 leakage-safe features. Absent candidates score zero in the target, and recent team selection and days since the last match let the model learn appearance risk directly. A decision layer then returns the best legal eleven by enumerating every feasible role-count split over sorted role lists, with a minimum-loss swap that guarantees both teams are represented. Every feature is computed from prior state only, under a hard data cutoff of 31 December 2025, and a step-by-step leakage audit is documented.

On 116,521 candidate rows from 3,577 matches in a temporal holdout, the model reaches an MAE of 19.478, a Spearman correlation of 0.564 and a top-11 capture ratio of 66.63%, compared with 28.992, 0.251 and 57.16% for a last-5 baseline. Removing the selection and recency features raises MAE to 21.938. The model artifact records an appearance ROC-AUC of 0.8860265.

1. Introduction and Problem Statement

1.1 Task definition

Before the toss, the system receives the match details and two full squads. It predicts every candidate's fantasy points and returns exactly one legal 11-player XI that maximises the sum of predicted points. The selection is made before the toss, so no information about the real playing XI, toss outcome or match conditions on the day may be used. The problem combines zero-inflated regression, within-match ranking and constrained optimization.

1.2 Why the problem is difficult

- **Availability uncertainty:** A squad contains more players than take the field. Selecting a player who does not play yields zero points, regardless of how strong that player is.
- **Hard structural constraints:** The XI must contain 1–4 wicket-keepers, 3–6 batters, 1–4 all-rounders and 3–6 bowlers, and must include players from both teams. A model that only ranks players cannot ensure this.
- **High-variance targets:** Single-match fantasy scores are heavy-tailed: the mean is 24.90 points but the median is 9.0, so rare explosive scores dominate squared-error metrics.
- **Temporal leakage risk:** Rolling statistics can silently include information from the match being predicted or from later matches, inflating offline results that then fail to transfer.

1.3 Contributions

- An **availability-aware training target**: absent candidates score zero, and recent selection and recency features let one model learn appearance risk.

- A **leakage-safe feature pipeline** in which every feature is written from prior state and state is updated only after the feature rows of a match are emitted.
- A **single-model point predictor**: histogram gradient boosting with MAE loss on 22 pre-match features.
- An **exact constrained selector** that enumerates feasible role-count splits over sorted role lists and guarantees 11 players, valid role ranges and both teams represented.
- A documented **audit trail** covering data scope, leakage controls, evaluation and limitations.

2. Data and Scoring

2.1 Data source

The only match source is the official Cricsheet JSON archive. The pipeline keeps T20 and IT20 matches from 2005-02-17 through 2025-12-31, producing 13,085 matches, 11,458 unique players and 424,084 historical candidate rows. Using a single ball-by-ball source keeps scoring consistent across every match and player.

2.2 Delivery-level scoring

Ball-level events are converted to the supplied fantasy rules, and Super Overs are ignored. Fantasy points are therefore computed directly from delivery-level rules rather than taken from any pre-aggregated summary, so any player's score in any match can be recomputed from the raw deliveries. A candidate who is not in the official XI is assigned zero points, which makes availability part of the target.

2.3 Candidate pools

Cricsheet does not publish full historical squads, so each team's five most recent XIs approximate its pre-toss candidate pool. This adds realistic zero labels for rotation risk without pretending that every observed player was certain to appear. Overall, 64.4% of candidate rows are players who played.

2.4 Temporal scope

A hard data cutoff of **31 December 2025** applies to the whole project. No record dated after December 2025 is used for training, validation, feature construction or testing.

Item	Specification
Source	Cricsheet ball-by-ball JSON archive, T20 and IT20 matches
Coverage	13,085 matches, 11,458 players, 424,084 candidate rows (2005-02-17 to 2025-12-31)
Target	Fantasy points from delivery-level rules; candidates not in the official XI score 0
Target shape	Mean 24.90, median 9.0, 64.4% played rate; right-skewed
Candidates	Each team's five most recent XIs
Cutoff	Hard cutoff 2025-12-31 for every stage of the pipeline
Ordering	All processing follows match date order

3. System Architecture

The system is organised as five stages. The guiding principle is a strict division of labour: **the statistical model ranks; the decision layer guarantees legality.** This keeps the learned component free to focus on estimation quality, while feasibility is handled by an exact enumeration that cannot produce an illegal XI.

Stage	Input	Output	Key property
1. Acquire	Cricsheet JSON files	Parsed matches and rosters	Single source of truth
2. Score	Delivery-level records	Fantasy points per player-match	Rule-based, reproducible
3. Featurize	Prior state only	Feature rows per candidate	Leakage-safe by construction
4. Point model	Feature rows	Predicted points per candidate	Gradient boosting, MAE loss
5. Optimize	Predicted points, roles, teams	One legal 11-player XI	Exact role-count enumeration

4. Leakage Audit

Leakage control is treated as a first-class part of the methodology rather than a final check. This section states the threat model, the controls that address each threat and the temporal protocol.

4.1 Threat model

- **Future-into-past leakage:** a feature for a match includes statistics from that match or later matches.
- **Same-date contamination:** two matches on the same date see each other's results through updated state.
- **Target overlap:** a feature is derived from the same deliveries that define the target.
- **Official-XI information:** using the official XI as a feature or candidate signal.
- **Identity collisions:** mixing men's and women's national-team histories under the same team name.
- **Split contamination:** random row splits, or training data reaching into the evaluation period, or matches dated after the cutoff entering training.

4.2 Sequential state protocol

Features are produced by a fixed five-step procedure in which the state is always read before it is updated.

Step	Purpose
1. Sort matches by date	Fixes an unambiguous meaning of "prior"
2. Read prior state	Each row sees only history from before its own match date
3. Write feature	Feature values are written from that snapshot
4. Finish same-date rows	All matches on one date are featurized before any state changes

Step	Purpose
5. Update state as batch	State moves forward once per date, so same-date matches cannot see each other

4.3 Control matrix

Risk	Control	Stage
Future-into-past	Prior-state-only features; chronological ordering	3. Featurize
Same-date contamination	Batch state update after all same-date rows	3. Featurize
Target overlap	Features drawn from state that excludes the current match	2–3
Official-XI information	Toss, official-XI flags, scorecard results and target-match events are never features; candidates come from each team's five most recent XIs	3. Featurize
Identity collisions	Team-history state is keyed by gender	3. Featurize
Venue and opponent aggregates	Computed chronologically, never over the full dataset	3. Featurize
Split contamination	Temporal holdout; date filter in iter_matches; train.py refuses any train end after 2025-12-31	All stages
Illegal outputs	Selector guarantees legality and the XI is rechecked before the CSV is written	5. Optimize

4.4 Temporal protocol and integrity checks

Training rows precede 2024-01-01. Validation covers all matches from 2024-01-01 to 2025-12-31 (3,577 matches, 116,521 candidate rows). The final model artifact is trained through 2025-12-31 and is never given later matches. The raw archive may contain matches after the cutoff, but the loader filters by event date and the training command rejects any later end date; the latest processed match date is exactly 2025-12-31.

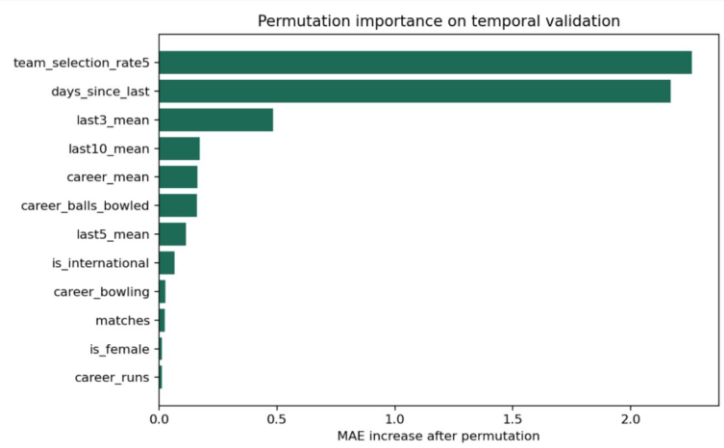
A manual audit found and fixed two problems: a gender collision in national-team histories and stale cross-team transfer candidates. The final data has zero duplicates on (match_id, team, player_id) and no missing target values.

5. Feature Engineering

Twenty-two features in seven groups describe each candidate. All are functions of prior state. The table below lists the groups and the signals each captures.

Group	Features	Rationale
Career	Runs, wickets, batting, bowling, fielding, matches	Long-run baseline of a player's ability and exposure
Recent Form	Mean points, last 3, 5 and 10	Form at short, medium and longer horizons
Form Shape	Last-5 standard deviation	Volatility of recent scores
Availability	Team selection rate (last 5), known candidate	How often the team has selected the player recently

Group	Features	Rationale
Schedule	Days since last match	Recency, rotation and selection likelihood
Team & Opponent	Team form, opponent points allowed	Strength of the team and the opposition
Context	Venue scoring level, team type, gender	Match setting; team history is keyed by gender

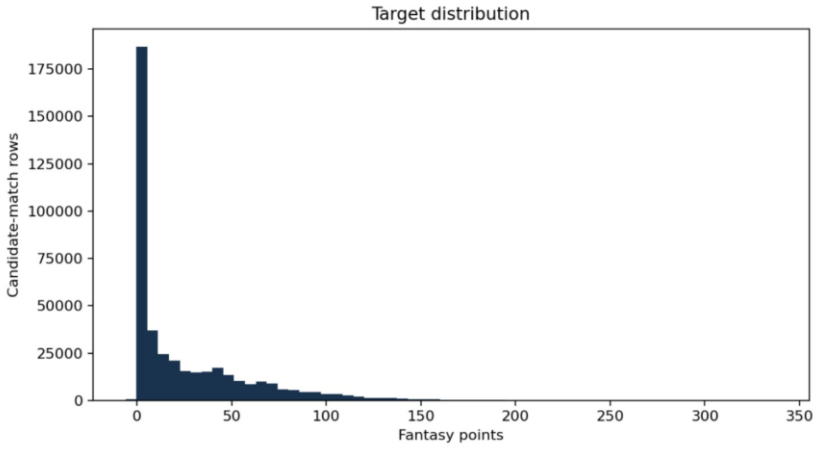


Multiple form windows (last 3, 5 and 10) let the model trade off responsiveness against stability, while the last-5 standard deviation summarises dispersion, which matters for heavy-tailed fantasy scores. The Availability and Schedule groups carry the appearance information that a single regressor needs on a zero-inflated target. In permutation importance, the most important features were **team_selection_rate5** and **days_since_last**.

6. Modelling

6.1 Availability-aware target

The supervised target is each candidate's realised fantasy points, and a candidate who is absent from the official XI receives zero. The target is therefore zero-inflated: 35.6% of candidate rows are zero by construction, with a long right tail from centuries and wicket hauls. Rather than fitting separate appearance and scoring models, the system gives one regressor the features that determine appearance risk (recent selection rate and recency) alongside form, career and context.



6.2 Point model

The point model is a histogram gradient boosting regressor:

$$\hat{y}_i = f_{HGB}(x_i)$$

Setting	Value
Loss	MAE
Iterations	250
Learning rate	0.06
Leaves	31
L2 regularization	2.0

- **MAE loss** is interpretable in fantasy points and is less dominated by rare explosive scores than squared error. It stabilises median-like predictions, at the cost of compressing high upside (Section 8.5).
- **Histogram gradient boosting** handles nonlinear interactions between availability, recent form and career role without large categorical encodings, and trains quickly on the 424,084 candidate rows.

6.3 Appearance signal

There is no separate appearance classifier in the scoring path. Appearance is carried by the recent team selection rate and days since last match, and the model artifact records an appearance ROC-AUC of **0.8860265**. A dedicated two-stage appearance and conditional-points model is listed as future work.

6.4 Final model and training protocol

The model artifact contains the fitted booster, the feature list, cutoff metadata, the registry-name map and all chronological state needed for future fixtures. New-name cold starts use conservative zero history, and fixture-provided roles override historical role inference. Models are trained chronologically under the hard cutoff of 31 December 2025.

7. Exact Constrained XI Selection

7.1 Formulation

Let candidate i have predicted points v_i and role r_i, and let x_i ∈ {0, 1} indicate selection. The selection problem is:

$$\text{maximise } \Sigma\, v_i \cdot x_i$$

- Total selected: Σ x_i = 11
- Role ranges: WK 1–4, BAT 3–6, AR 1–4, BOWL 3–6
- Team coverage: at least one player from each of the two teams

7.2 Why not a ranking cut-off

Taking the eleven highest predictions is fast but not safe. The top eleven may contain too many players of one role, too few of another, or players from only one side. Repairing an illegal selection by swapping players is a heuristic with no optimality guarantee, and the best legal XI can differ from the repaired top-eleven.

7.3 Role-count enumeration

Candidates are grouped by role and each role list is sorted by predicted points. The optimizer enumerates every feasible role-count split (w, b, a, o) with each count inside its range and w + b + a + o = 11. For a fixed split, the best choice is the top w wicket-keepers, top b batters, top a all-rounders and top o bowlers, because a player belongs to exactly one role list. The split with the highest total predicted points is returned. If the two teams are not both represented, a minimum-loss swap brings in the missing side.

Component	Definition
Inputs	Predicted points, roles (fixture roles override inferred roles) and teams
Enumeration	Every (WK, BAT, AR, BOWL) count split within the role ranges that sums to 11
Selection	Top-ranked players from each sorted role list for that split
Best split	The split with the highest total predicted points
Team coverage	Minimum-loss swap if both teams are not initially represented
Guarantees	11 total, role ranges, both teams
Post-check	The result is rechecked before the CSV is written

7.4 Complexity

Substituting each count minus its minimum gives non-negative counts that sum to 3, so there are C(6, 3) = 20 feasible role-count splits, regardless of squad size. The work per fixture is one sort per role list plus 20 sums of eleven values. With a pool of 30 candidates, brute-force enumeration would need C(30, 11) = 54,627,300 subsets.

7.5 Justification against alternatives

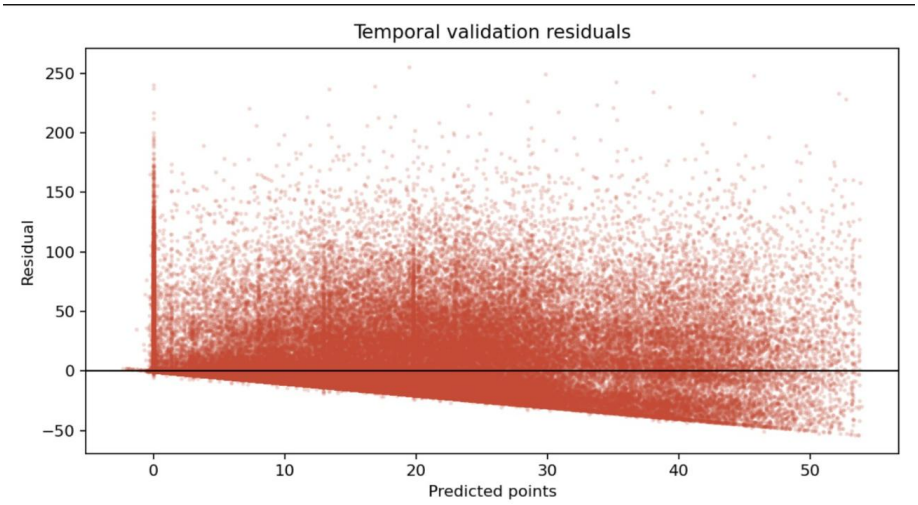
Strategy	Optimal	Always legal	Comment
Top-11 by value	No	No	Ignores role and team rules
Greedy with repair	No	Yes	Legal, but repair steps can discard value with no guarantee
Brute-force enumeration	Yes	Yes	Exact but grows combinatorially with pool size
Role-count enumeration (used)	Yes, for role constraints	Yes	20 splits, deterministic and easy to audit; team coverage by minimum-loss swap

Role-count enumeration was selected because it is exact for the role constraints, its cost is small and predictable, and its behaviour is fully deterministic, which supports the audit requirements of this project.

8. Experiments and Results

8.1 Evaluation protocol

Training rows precede 2024-01-01. Validation covers all matches from 2024-01-01 to 2025-12-31, a blind-like temporal holdout of 3,577 matches and 116,521 candidate rows. Every rolling statistic is generated by walking matches in date order, and a random row split is avoided because it would overstate performance. MAE is the primary calibration metric; RMSE highlights costly misses on explosive scores; Spearman correlation and top-11 capture measure the ranking decision. Capture equals the points earned by the predicted top 11 divided by the hindsight-oracle top 11, before role constraints. Top-11 overlap is the agreement between the predicted and oracle top 11.



8.2 Headline results

Metric	Value	Interpretation
Top-11 capture vs last-5 baseline	66.63% vs 57.16% (+9.47 pp)	Share of oracle top-11 points captured
MAE vs last-5 baseline	19.478 vs 28.992 (−32.8%)	Mean absolute error in fantasy points
Spearman correlation	0.564 vs 0.251	Ordering quality of candidates
Top-11 overlap	56.41% vs 48.10%	Agreement with the oracle top 11
MAE, played vs absent	27.14 vs 5.53	Error split by whether the candidate played
Appearance ROC-AUC	0.8860265	Recorded in the model artifact

8.3 Model comparison

The table compares the final booster with Extra Trees and with a rolling-mean baseline. Lower is better for MAE and RMSE; higher is better for Spearman, capture and overlap.

Model	MAE	RMSE	Spearman	Capture	Overlap
Last-5 mean	28.992	37.145	0.251	0.5716	0.4810
Extra Trees	22.190	30.785	0.550	0.6654	0.5611

Model	MAE	RMSE	Spearman	Capture	Overlap
Histogram gradient boosting	19.478	32.292	0.564	0.6663	0.5641

The booster wins MAE, Spearman, capture and overlap. Extra Trees wins RMSE, suggesting fewer very large errors, but its overall selection quality is slightly lower, so the booster is the final model. Against the baseline, the booster improves MAE by 32.8% and capture by 9.47 percentage points.

8.4 Ablation

Features	MAE	Spearman	Capture
Full 22-feature model	19.478	0.564	0.6663
Remove recent selection and days since last match	21.938	0.334	0.5972
Form-only seven features	22.251	0.303	0.5902

Removing the selection and recency features raises MAE from 19.478 to 21.938 and lowers capture from 66.63% to 59.72%. A form-only model reaches MAE 22.251 and capture 59.02%. Appearance information therefore contributes more than additional form horizons alone.

8.5 Error analysis

Candidate rows are split by whether the player played and by prior experience. Players who played account for 74,774 of 116,521 validation rows (64.2%).

Played	Experience band	Rows	MAE	Mean actual	Mean predicted
Yes	debut/unknown	2,376	22.35	28.02	16.83
Yes	1-4	7,153	23.08	29.29	11.71
Yes	5-14	12,616	24.78	34.13	17.35
Yes	15+	52,629	28.47	41.54	23.01
No	1-4	7,993	2.26	0.00	2.26
No	5-14	8,888	3.94	0.00	3.93
No	15+	24,866	7.15	0.00	7.15

The model predicts absences accurately but compresses the upside of players who appear: MAE is 27.14 for played candidates and 5.53 for absent ones. Established players are hardest, because they combine high expectation with high match-to-match variance; in the 15+ match group the mean actual score is 41.54 and the mean prediction is 23.01. MAE training stabilises median-like predictions but compresses high upside.

8.6 Inferences

Lifting top-11 capture from 57.16% to 66.63% shows that learned features add substantial value beyond recent form alone, and the ablation shows that availability features provide the largest measurable gain. An MAE of 27.14 on played rows is large in absolute terms, consistent with the high

intrinsic variance of single-match fantasy scores; the selector therefore relies on the relative ordering of predictions rather than on exact point forecasts.

9. Model Explainability

9.1 Interpretability by design

The model uses 22 named pre-match features in seven groups, each with a clear cricket meaning. Because availability is carried by two explicit features (team selection rate and days since last match), a reviewer can see whether a candidate is ranked highly because he is likely to play, because his form and career record are strong, or both.

9.2 Feature importance

Permutation importance on the validation set measures how much MAE rises when a feature is shuffled. **team_selection_rate5** (+2.26 MAE) and **days_since_last** (+2.17) rank far above every other feature; the next is last3_mean (+0.48), and recent form is the next most important family. The two leading features belong to the Availability and Schedule groups, indicating that how often and how recently a player has been selected carries the strongest signal. The standard deviations across repeats are small (0.04 and 0.06 for the top two), so the ranking is stable.

Feature	MAE increase (mean)	Std. deviation
team_selection_rate5	2.260	0.036
days_since_last	2.170	0.057
last3_mean	0.484	0.017
last10_mean	0.174	0.012
career_mean	0.163	0.011
career_balls_bowled	0.162	0.014
last5_mean	0.116	0.013
is_international	0.065	0.005

Several features contribute essentially nothing (last5_std and candidate_known at 0.000; venue_points_mean is marginally negative).

9.3 Selector transparency

The selection step is deterministic. Given the same predictions, roles and teams, it returns the same XI, and every constraint it enforces is explicit. The winning role-count split and the sorted role lists give a complete record of why a player was included or excluded.

9.4 Recommended diagnostics

- Report predicted points alongside team selection rate and days since last match for every selected player.
- Report per-group feature contribution for a sample of selected and rejected candidates.
- Track error separately for players who played and for absent candidates.

10. Limitations

Limitation	Effect	Planned response
Ceiling scores underpredicted	MAE training compresses upside; played-candidate MAE is 27.14 against 5.53 for absent ones	Two-stage model, or quantile or ranking objective
Inferred roles can fail	Historical roles are approximate; a candidate may land in the wrong role group	Fixture-provided roles already override; authoritative role source
Candidate pools are approximations	Cricsheet does not publish full historical squads, so the five most recent XIs stand in for them	Use announced squads where available
Direct-hit run-outs	Cannot be distinguished from other run-outs in the data	None planned
Cold-start players	Little history means little signal	Conservative zero history

11. Next Iterations

- **Two-stage appearance and conditional-points model** to separate the probability of playing from the score given an appearance.
- **Authoritative roles** to replace inferred historical roles and remove a source of constraint errors.
- **Conditional quantile or ranking objective** to recover the upside that MAE training compresses.

12. Reproducibility and Auditability

- **Deterministic scoring:** targets are recomputable from raw deliveries.
- **Deterministic selection:** the optimizer returns identical output for identical input.
- **Fixed temporal boundary:** the 31 December 2025 cutoff applies to every stage and is enforced in code.
- **Output recheck:** the final XI is validated against every constraint before the CSV is written.
- **Stored artifacts:** the archive hash, experiment tables, internal predictions, plots and final model are kept in the repository, and unit tests cover scoring, optimizer constraints and cutoff behaviour.

13. Conclusion

Final Model v3 treats fantasy team selection as a prediction problem under availability uncertainty and hard structural constraints. A single histogram gradient boosting model with availability-aware features lifts top-11 capture from 57.16% to 66.63% and cuts MAE by 32.8% over a last-5 baseline on a blind temporal holdout, and an exact role-count enumeration delegates legality to a deterministic optimizer. A strict read-before-update protocol and a hard cutoff at 31 December 2025 keep the pipeline leakage-safe and auditable. The identified limitations, chiefly compression of ceiling scores and approximate roles and candidate pools, define a clear agenda for the next iteration.

Constraint Summary

Rule	Minimum	Maximum
Wicket-keepers (WK)	1	4
Batters (BAT)	3	6
All-rounders (AR)	1	4
Bowlers (BOWL)	3	6
Total players	11	11
Teams represented	Both	Both

Notation

Symbol	Meaning
ŷ_i or v_i	Predicted fantasy points for candidate i, used by the selector
x_i	Binary selection indicator for candidate i
Capture	Points earned by the predicted top 11 divided by the hindsight-oracle top 11, before role constraints
Overlap	Agreement between the predicted and oracle top 11
MAE	Mean absolute error
RMSE	Root mean squared error
Spearman	Rank correlation between predicted and actual points
ROC-AUC	Area under the receiver operating characteristic curve